Revisiting RLMRec: User-Profile Alignment, Early Stopping, and Reproducibility

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Abstract. We reproduce selected released-code behavior and study the user loss in RLMRec-Con with a LightGCN backbone on Amazon-book, Yelp and Steam. Removing it changes mean test Recall@20 by -0.00027, -0.00015 and -0.00139 over five paired seeds. Both full and item-only objectives outperform the collaborative baseline; only Steam provides paired t-test evidence of a user-term benefit after multiplicity adjustment. The other intervals do not establish equivalence. A diagnostic explains a much larger apparent effect on Yelp: longer patience lets the full model recover from about 0.038 to 0.116 validation Recall@20. Fixed-seed repeats also vary on the scale of the small Amazon mean contrast. This objective-level ablation identifies a conditional user-term benefit and shows how premature stopping can exaggerate an apparent benefit from its removal.

Keywords: Recommendation · Language models · Reproducibility · Ablation studies

1 Introduction

RLMRec aligns collaborative representations with generated user and item profiles [1]. An improvement from this combined objective does not establish how much each alignment term contributes. Separating their effects matters when choosing whether a recommender needs a user-profile generation pipeline in addition to item semantics.

The original RLMRec ablation (Section 4.3) examines shuffled semantic representations and alternative text encoders [1]. It does not report our intervention: deleting the user-alignment loss while keeping the item losses and released embeddings fixed. Corrupting semantic targets and removing one loss answer different questions. The former tests the relevance of semantic correspondence; the latter tests a term's marginal contribution.

We test this intervention on three benchmarks and diagnose a high-weight Yelp failure using longer patience. Fixed-seed repeats check numerical repeatability. The contribution is an objective ablation that separates a conditional user-term benefit from a stopping artifact. This retrospective study does not propose a new state-of-the-art model.

2 Context and Compared Objectives

LightGCN propagates ID embeddings over the interaction graph [3]. RLMRec adds alignment to generated profiles [1]. AlphaRec already shows that item lan-

guage representations can support recommendation and examines alternative RLMRec inputs [4]; our intervention instead keeps the released inputs fixed and deletes one alignment term. An earlier FREEDOM-inspired graph recipe [2], which changes several components at once, is retained only in the supporting artifact.

Let a training batch contain B triples (u, i, j), with observed item i and an item j sampled outside that user's training history. With collaborative embeddings h and score $\hat{y}_{ui} = h_u^\top h_i$, the averaged BPR loss is

$$\overline{\mathcal{L}}_{\mathrm{BPR}} = \frac{1}{B} \sum_{(u,i,j)} \mathrm{softplus}(\hat{y}_{uj} - \hat{y}_{ui}). \tag{1}$$

A shared two-layer MLP g maps frozen 1536-dimensional semantic vectors into the collaborative dimension. Write $\overline{\mathcal{L}}_u$, $\overline{\mathcal{L}}_{i^+}$ and $\overline{\mathcal{L}}_{i^-}$ for the batch-averaged cosine InfoNCE losses on users, positive items and negative items, respectively. Following the released code, each user target is contrasted against all projected user profiles; the two item losses use their respective in-batch item targets. Repeated items are retained in the batch. Then the full and item-only objectives are

$$\mathcal{L}_{\mathrm{full}} = \overline{\mathcal{L}}_{\mathrm{BPR}} + \alpha(\overline{\mathcal{L}}_u + \overline{\mathcal{L}}_{i^+} + \overline{\mathcal{L}}_{i^-}) + \beta\|\Theta\|_2^2, \tag{2}$$

$$\mathcal{L}_{\mathrm{item}} = \overline{\mathcal{L}}_{\mathrm{BPR}} + \alpha(\overline{\mathcal{L}}_{i^+} + \overline{\mathcal{L}}_{i^-}) + \beta\|\Theta\|_2^2. \tag{3}$$

Deleting $\overline{\mathcal{L}}_u$ leaves the other coefficients unchanged. It reduces total alignment pressure and changes the shared projector's supervision, so this is not a comparison at matched total regularization. Both RLMRec variants retain the released sum of layers 0 through 3, dropout without survivor rescaling and squared parameter regularization. Cached embeddings are invalidated before evaluation and after checkpoint restoration.

3 Reproduction Scope and Process

Why this system. RLMRec provides a concrete instance of a widely applicable design question: how should generated semantic profiles supervise collaborative recommendation? Its released interactions, semantic vectors and LightGCN implementation allow us to hold the inputs fixed while separating the user and item alignment terms. We focus on RLMRec-Con with this backbone. We do not reproduce every backbone, augmentation variant or original-paper experiment, and do not claim numerical equality with its published tables. The contribution is a released-code reproduction with an objective-level extension and a diagnosis of implementation and stopping choices that affect its interpretation.

Questions and evidence. We ask whether removing only user alignment preserves the measured semantic gain, whether that contrast is stable across three released benchmarks, and whether an apparent large loss difference survives a

longer stopping budget. A fourth question concerns execution repeatability at a fixed seed. These questions require different evidence: five paired seeds for main model contrasts, validation-only runs for the stopping diagnosis, and repeated executions of one fixed setting for numerical repeatability. We keep those samples separate rather than pooling them as additional independent seeds.

Reconstructing the implementation. The local audit identifies the RLMRec release at commit `22413752246de3dee8ab0d509f7f7a8889080f95`. The important reconstruction choices are the released sum readout, edge dropout without survivor rescaling, global squared parameter regularization, and the distinct user and item contrastive candidate sets. Earlier local LightGCN variants used different readouts or regularization and are excluded from the main comparison. Agreement on a model name is therefore insufficient to establish that two experiments use the same objective or propagation operator.

The local implementation invalidates its embedding cache before evaluation and after restoring a checkpoint. Otherwise evaluation may use embeddings computed from earlier weights. This is a deliberate implementation correction, not an additional model component; we do not have a controlled benchmark estimate of its isolated effect on accuracy. The tests cover the retained objective, model wiring, graph operators and cache behavior. Passing them supports these implementation properties rather than establishing equality with every upstream training outcome.

Changes and acquisition limits. We reuse the supplied semantic vectors rather than regenerating profiles. Training uses Apple MPS, with a gather/scatter implementation of sparse propagation. Dataset-specific batches and stopping budgets are stated below; they are changes in experimental setup that limit a strict replication claim. The evidence package records public-source and local-file audits, but no upstream-author correspondence. We cannot infer that undocumented details were confirmed by the original authors. Nor have we byte-compared these local inputs with a newly downloaded complete release.

What can be reproduced from the package. The artifact supports three progressively stronger checks. First, a reader can verify file fingerprints and recalculate every reported main contrast from frozen JSON results without raw data or accelerator hardware. Second, with the released inputs installed, the reader can verify split disjointness, array shapes and input fingerprints. Third, the training entry point reruns a selected manifest configuration into a separate output directory. A two-epoch smoke run exercises the pipeline but is not a reproduction of the scientific result. Identical seeds do not guarantee identical MPS outcomes.

The main manifest fixes 60 runs: four models, three datasets and five seeds. This paper reports 45 of them, omitting the graph-recipe comparison from its main tables while retaining it in the multiplicity family and artifact. The package also includes all twelve weight-grid cells and twelve diagnostic executions. Only the weight grid has a fully preserved pre-run source snapshot; source digests for earlier and later runs do not recover missing source bytes. This distinction

prevents table reproducibility from being mistaken for a complete reconstruction of every historical execution.

4 Experimental Design

Inputs and ranking. We reuse RLMRec's released interactions and semantic embeddings. Amazon-book contains 11,000 users, 9,332 items and 120,464 training, 40,290 validation and 40,106 test interactions. Yelp contains 11,091 users, 11,010 items and 166,620 training, 55,479 validation and 55,436 test interactions. Steam contains 23,310 users, 5,237 items and 316,190/104,897/104,835 training/validation/test interactions. We verified that the split matrices have disjoint nonzero pairs and that the semantic inputs have the expected dimensions and finite values. Collaborative adjacency uses training data only. All-ranking evaluation masks training positives; validation positives remain eligible test candidates. Recall and NDCG are macro-averaged over users with positives in the evaluated split.

We cannot verify a complete split-filtered generation history for every released semantic profile. The upstream examples use GPT-3.5-turbo and text-embedding-ada-002, but examples are not authenticated generation logs for the full release. Our results are conditional on these fixed semantic inputs; disjoint interaction splits alone do not certify absence of information contamination in upstream text generation.

Configurations and selection. All runs use Apple MPS. Compared models use 32-dimensional collaborative embeddings, three collaborative layers, Adam with learning rate 10^{-3} and no optimizer weight decay. Full and item-only RLMRec share $\alpha = 0.01$, temperature 0.2, keep rate 0.8 and $\beta = 10^{-7}$. Batch sizes are 1024 for Amazon and 4096 for Yelp and Steam, fixed within each dataset. Consequently, cross-dataset behavior is not an isolated effect of dataset identity. The semantic models have 1,880,752 trainable parameters on Amazon and 1,937,360 on Yelp.

We evaluate validation Recall@20 every three epochs and restore its best checkpoint before a single final test evaluation per run. Patience is five non-improving evaluations for Amazon and Yelp and 25 for Steam. Steam is a later extension with a longer stopping budget, not an identical-protocol replication across datasets. The five paired seeds are 1, 7, 13, 42 and 2024. All 45 runs in the three-model comparison stopped early rather than hitting a time or epoch cap. Existing development and test results informed this retrospective study; its framing is not preregistered.

Uncertainty and sensitivity. We report sample standard deviations and paired 95% Student-t intervals across the five seeds, with paired t-tests and exhaustive two-sided sign-flip checks. Holm adjustment covers twelve contrasts per metric: full and item-only versus the baseline, and item-only and the archived graph recipe versus full, on each of three datasets. Intervals are pointwise, not

Table 1. Test metrics at 20, mean ± sample SD over five seeds. All model comparisons are matched within a dataset.

Dataset	Model	Recall@20	NDCG@20
Amazon	LightGCN	0.1386 ± 0.0012	0.0849 ± 0.0004
Amazon	Full RLMRec	0.1465 ± 0.0023	0.0894 ± 0.0010
Amazon	Item-only RLMRec	0.1462 ± 0.0013	0.0895 ± 0.0006
Yelp	LightGCN	0.1130 ± 0.0026	0.0713 ± 0.0017
Yelp	Full RLMRec	0.1227 ± 0.0015	0.0769 ± 0.0010
Yelp	Item-only RLMRec	0.1225 ± 0.0016	0.0773 ± 0.0005
Steam	LightGCN	0.1319 ± 0.0007	0.0834 ± 0.0005
Steam	Full RLMRec	0.1424 ± 0.0009	0.0903 ± 0.0005
Steam	Item-only RLMRec	0.1410 ± 0.0006	0.0893 ± 0.0002

multiplicity-adjusted. The family includes the archived comparisons even though they are omitted from the main tables. Five seeds quantify run variation on fixed inputs, not uncertainty across populations; no equivalence margin was selected.

The original validation-only grid runs full and item-only RLMRec at $\alpha \in \{0.003, 0.01, 0.03\}$ on development seed 2023 for Amazon and Yelp. All twelve cells are retained. To diagnose the high-weight Yelp result, four additional runs use $\alpha = 0.03$, patience 20 and seeds 2023 and 2025, with other settings unchanged. They evaluate no test metrics. This follow-up is motivated by the observed failure and is not a new unbiased test comparison.

Separately, each RLMRec variant was executed four times at the fixed Amazon setting $\alpha = 0.01$, seed 42 and patience five. Each repeat reports its validation-selected test score. These eight executions measure repeatability; they are not added to the five-seed sample or used to select a configuration.

5 Results

Marginal contribution of user alignment. Full and item-only RLMRec improve on the collaborative baseline on every dataset (Table 1); all six paired t comparisons survive Holm adjustment, with adjusted $p \leq 0.020$. The item-only gains in Recall@20 are 0.00760, 0.00959 and 0.00907 on Amazon, Yelp and Steam. These gains show that a semantic objective without the user term can be effective in this implementation. They do not assign all of full RLMRec's benefit to item information: the objectives share a projector and differ in total loss weight.

For Amazon and Yelp, Table 2 reports changes of -0.000266 and -0.000151. Their pointwise intervals span zero but admit losses as large as approximately 1.25% and 2.08% of full RLMRec's mean; neither establishes practical equivalence. On Steam the change is -0.001390 (about -0.98% relative), with a pointwise interval excluding zero and Holm-adjusted paired t value $p = 0.0419$. Thus the data include a setting where user alignment helps. With only five pairs, the exact sign-flip test cannot attain a two-sided value below 0.0625; the Steam in-

Table 2. Item-only minus full RLMRec, test Recall@20. Pointwise 95% paired t intervals; p_h is Holm-adjusted across twelve contrasts per metric and p_s is the exact two-sided sign-flip value.

Dataset	Difference	95% t interval	p_h	p_s
Amazon	−0.00027	[−0.00182, +0.00129]	1.0000	0.6875
Yelp	−0.00015	[−0.00255, +0.00224]	1.0000	0.7500
Steam	−0.00139	[−0.00231, −0.00047]	0.0419	0.0625

Table 3. Yelp $\alpha = 0.03$, patience 20. Validation Recall@20 and best epoch indices. These runs never evaluate the test set.

Seed	Full	Item-only	Full epoch	Item epoch
2023	0.11587	0.11866	84	66
2025	0.11605	0.11991	72	63

ference depends on the parametric test. The inference is restricted to the tested Steam protocol.

Alignment strength. The artifact reports every original grid cell. On Amazon, the item-only minus full difference changes from −0.00184 at $\alpha = 0.003$ to +0.00020 at 0.01 and +0.00359 at 0.03. On Yelp the models are close at the two smaller weights. At 0.03, full RLMRec selects epoch 0 and stops after 16 training epochs, while item-only reaches 0.11889. This large gap motivates the stopping diagnostic below. The grid does not support a weight-independent claim about the user loss or provide test performance for newly tuned models.

5.1 Diagnosing early stopping

Under patience 20, full RLMRec at Yelp $\alpha = 0.03$ reaches validation Recall@20 of 0.11587 and 0.11605 on seeds 2023 and 2025. Item-only reaches 0.11866 and 0.11991 (Table 3). The original seed-2023 short-patience gap was 0.08085; the long-patience follow-up gap is 0.00280. Full RLMRec still trails, but the severe apparent collapse does not persist.

The trajectories explain why (Figure 1). Both full-model runs initially fall below their epoch-0 validation values. Replaying the five-check stopping rule on their logged validation values would stop both at epoch index 15, before recovery; with more patience, their best epochs are 84 and 72. The logs round Recall@20 to four decimals, so replay is an approximation, although these early deficits are much larger than rounding. This within-trajectory check avoids assuming that the original and repeated MPS runs are bitwise identical. It supports premature stopping as the cause of the severe low-score outcome, without identifying its optimization cause.

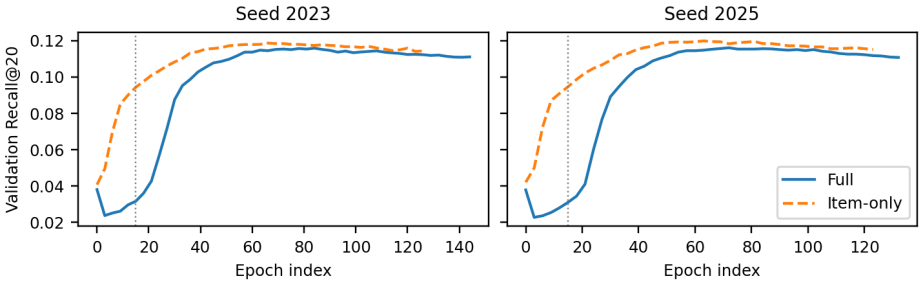


Fig. 1. Long-patience validation trajectories. Dotted lines mark epoch 15, where the five-check rule would stop each full-model trajectory.

5.2 Fixed-seed repeatability

The four Amazon repeats have test Recall@20 SD 0.000536 for full RLMRec and 0.000344 for item-only, with ranges 0.001090 and 0.000810. Full-model selected epoch indices are 66, 78, 66 and 72; item-only selects 66 in all four executions. Execution variation is therefore present even when checkpoint epoch is unchanged. Both SDs exceed the absolute Amazon five-seed mean contrast of 0.000266.

This observation cautions against interpreting the small Amazon point estimate precisely. It does not invalidate pairing, establish a universal detection threshold, or prove that the observed variation is caused solely by MPS scatter operations. Four repeats at one seed cannot decompose execution and initialization variance or be transferred to Yelp and Steam. The primary uncertainty estimates remain the paired seed intervals.

6 Implications and Limitations

The combined evidence supports a specific practical conclusion: evaluate the user term separately and inspect stopping trajectories before interpreting a large ablation gap. Semantic augmentation helps without user alignment in all three tested settings, but Steam shows a measurable incremental user-term benefit under the paired t analysis. The high-weight Yelp result demonstrates how stopping can greatly exaggerate that benefit's apparent reversal.

Our scope is one backbone, three fixed benchmarks and released semantic inputs with incompletely verified generation provenance. Dataset-specific batch sizes and patience prevent attributing all differences to dataset identity. The diagnostics neither optimize model performance nor establish equivalence. Removing a loss also changes shared-projector supervision and total alignment pressure.

Historical results and logs carry source and data digests, but complete source bytes are not recovered for every run, including the follow-up diagnostics. The original weight grid has a preserved pre-run snapshot. The artifact distinguishes these evidence levels and retains all scores, checksummed manifests, current code

and table-generation scripts. Removing user alignment still leaves collaborative user histories and potentially user-authored item reviews in use. We establish no new-user cold-start, privacy or profile-generation-cost benefit.

Artifact access. The software artifact supplies the implementation, result manifests, archived controls and analysis scripts. It also contains manuscript sources and the source snapshot for the weight grid. The README separates table regeneration, input checks and training. Raw datasets and semantic arrays must be obtained from RLMRec. The package manifest records the exact bytes supplied with these manuscripts.

AI assistance. OpenAI Codex assisted with study framing, code and statistical checks, execution of the validation-only sensitivity grid, literature searches, and manuscript drafting. Reported numerical results come from the archived training outputs and analysis scripts. This assistance does not replace author review or responsibility for the work.

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